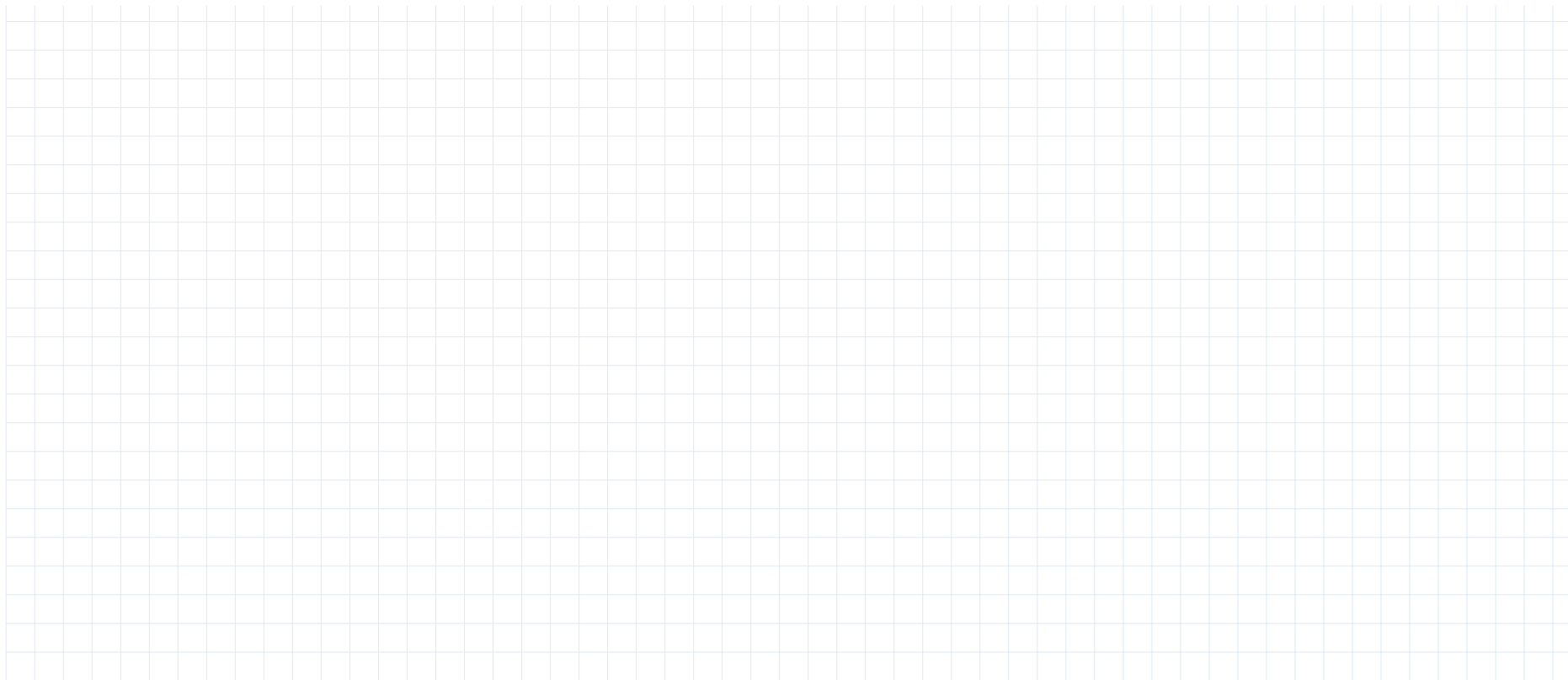


The sine graph

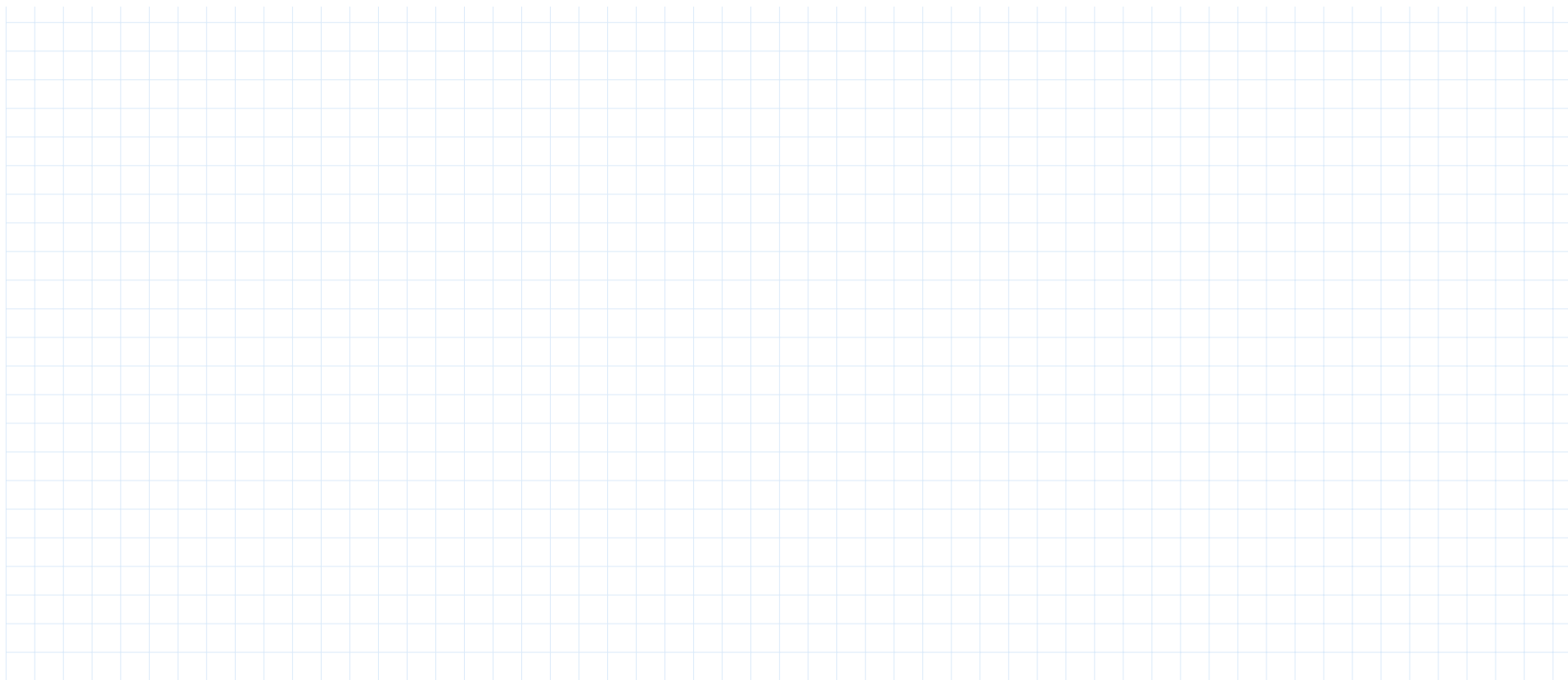
The graph of $y = \sin x$ repeats every 2π . Its values stay between -1 and 1 .

Example. Find the values at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$, and use these five points to sketch one cycle. State the domain, range, and period.



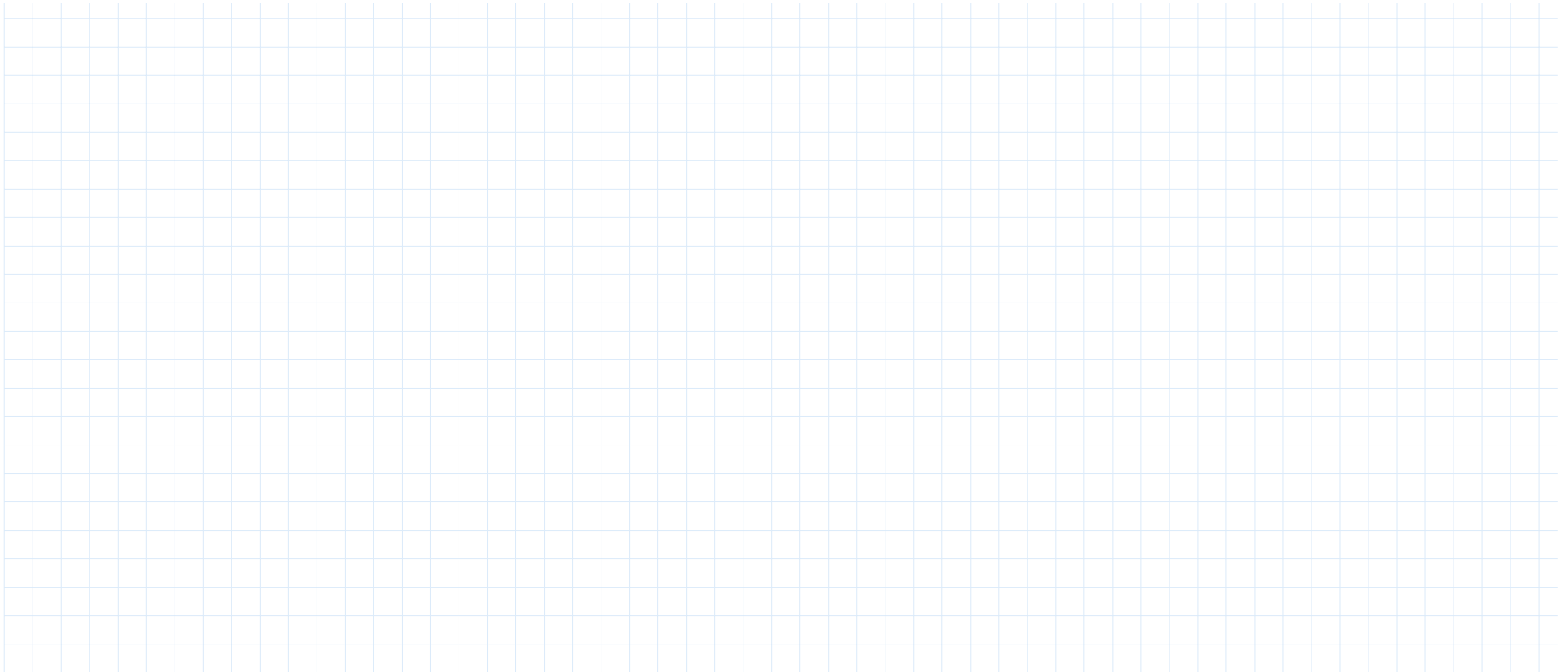
The cosine graph

Example. Sketch $y = \cos x$ for $0 \leq x \leq 2\pi$ using its five key points. State the domain, range, period, and y -intercept. How does its starting value differ from sine?



For $y = A \sin(Bx) + D$ or $y = A \cos(Bx) + D$, amplitude is $|A|$, period is $\frac{2\pi}{|B|}$, and the midline is $y = D$.

Example. For $y = -3 \sin(2x) + 1$, find the amplitude, period, midline, and range. Sketch one cycle using five points.

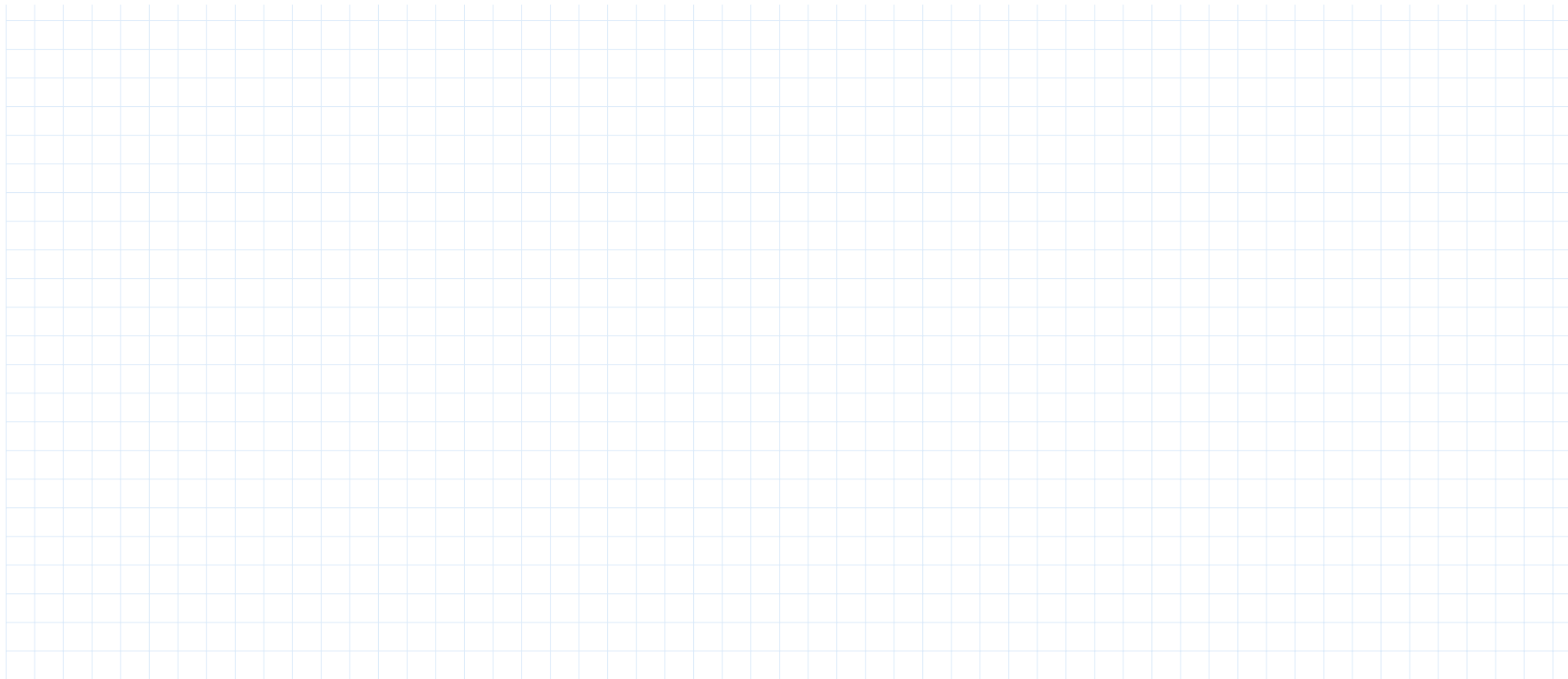


A stretched cosine cycle

Example. Sketch one cycle of

$$y = 2 \cos\left(\frac{x}{2}\right) - 1.$$

Label the five key points. State the amplitude, period, midline, domain, and range.



Writing a sinusoidal equation

Example. A sinusoidal graph has maximum value 5, minimum value -1 , and period 6. At $x = 0$, it crosses its midline while increasing.

Write an equation for the graph and verify the maximum and minimum values.

